

1. What is Enum in C? Give an example.

Enumeration (or enum) is a way of creating user defined data type in C. It is mainly used to assign names to integral constants. Enumeration data type consists of named integer constants as a list. It start with 0 (zero) by default and value is incremented by 1 for the sequential identifiers in the list.

Syntax

enum identifier {enumerator-list};

Egs –

```
enum month { Jan, Feb, Mar };
```

Jan, Feb and Mar variables will be assigned to 0, 1 and 2 respectively by default

```
enum month { Jan = 1, Feb, Mar };
```

Feb and Mar variables will be assigned to 2 and 3 respectively by default

3. What will be the output of the following C code?

```
#include <stdio.h>
enum example {a = 1, b, c};
enum example example1 = 2;
enum example answer()
{
    return example1;
}
int main()
{
    (answer() == a)? printf("yes"): printf("no");
    return 0;
}
```

Ans- no

Explanation: In the code shown above, the value of example1 is returned by the function answer. The ternary statement prints yes if this value is equal to

that of 'a' and no if the value is not equal to that of 'a'. Since the value of 'a' is 1 and that returned by the function is 2, therefore no is printed.

6. What is the benefit of using an enum rather than a #define constant?

The use of an enumeration constant (enum) has many advantages over using the traditional symbolic constant style of #define. These advantages include a lower maintenance requirement, improved program readability, and better debugging capability.

- 1) The first advantage is that enumerated constants are generated automatically by the compiler. Conversely, symbolic constants must be manually assigned values by the programmer.
- 2) Another advantage of using the enumeration constant method is that the programs are more readable and thus can be understood better by others.
- 3) A third advantage to using enumeration constants is that some symbolic debuggers can print the value of an enumeration constant. Conversely, most symbolic debuggers cannot print the value of a symbolic constant. This can be an enormous help in debugging user program, because if your program is stopped at a line that uses an enum, it can simply inspect that constant and instantly know its value. On the other hand, because most debuggers cannot print #define values, user would most likely have to search for that value by manually looking it up in a header file.